

PETROPAVLOVSK-KAMCHATSKY, Russian Federation

WMO: 325400

Lat: **53.083N**Lon: **158.583E**Elev: **275**StdP: **14.55**Time Zone: **12.00 (E12)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.2	-4.0	-18.2	2.1	-4.3	-13.5	2.7	-1.4	31.3	32.9	26.9	30.2	8.6	0	0.733

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.1	76.9	62.4	73.2	60.5	69.6	59.2	64.7	73.6	62.1	70.3	59.9	67.3	9.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.8	80.4	69.5	58.7	74.6	66.2	56.8	69.6	63.5	29.8	73.5	28.0	70.5	26.4	67.4	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 24.2	20.6	17.5		-12.6	83.4	4.6	4.0	-15.9	86.2	-18.7	88.5	-21.3	90.8	-24.6	93.6	
(5)				-13.2	68.5	4.5	3.3	-16.5	70.9	-19.1	72.8	-21.7	74.7	-25.0	77.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.0	14.8	16.6	22.8	32.1	41.9	51.5	58.0	58.4	51.2	39.7	26.6	16.9
	DBStd	17.19	10.04	9.19	7.51	5.35	5.05	6.10	5.51	4.21	5.09	5.46	7.09	9.14
	HDD50	5814	1090	936	843	538	258	51	4	1	44	320	702	1027
	HDD65	10611	1555	1356	1308	988	716	409	227	209	415	784	1152	1492
	CDD50	698	0	0	0	0	8	95	252	262	80	1	0	0
	CDD65	18	0	0	0	0	0	2	11	5	0	0	0	0
	CDH74	246	0	0	0	0	1	56	128	57	4	0	0	0
	CDH80	31	0	0	0	0	0	9	19	3	0	0	0	0
Wind	WSAvg	7.6	8.2	8.5	8.2	7.7	7.7	7.4	6.5	6.2	6.9	7.4	8.0	8.6
Precipitation	PrecAvg	39.8	3.1	2.5	3.0	3.1	2.2	2.6	2.6	4.0	3.8	5.5	4.6	3.3
	PrecMax	52.4	6.6	9.6	7.0	6.8	4.4	6.9	6.4	8.3	6.3	14.8	14.4	12.4
	PrecMin	28.3	0.5	0.4	0.6	0.8	0.7	0.6	0.2	1.2	0.8	1.5	0.9	0.7
	PrecStd	7.0	1.7	1.9	1.5	1.7	0.9	1.6	1.4	1.9	1.4	3.0	3.0	2.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.8	37.8	39.3	51.4	66.6	80.5	82.6	79.1	73.1	58.6	44.8	39.5
		MCWB	33.7	33.2	33.2	41.1	50.6	61.2	64.8	65.6	58.0	46.6	41.8	35.1
	2%	DB	35.5	35.5	37.3	46.2	62.2	73.5	77.3	75.2	68.0	54.6	41.4	37.2
		MCWB	31.6	31.6	32.2	37.8	48.9	57.4	63.3	62.2	55.8	45.3	38.1	33.4
	5%	DB	33.5	32.7	35.4	42.8	57.3	69.4	73.4	71.7	64.4	51.7	39.4	34.1
		MCWB	30.2	29.6	31.2	35.7	46.2	56.6	61.8	60.8	53.6	43.4	36.1	31.6
	10%	DB	30.9	30.5	33.6	39.6	52.2	64.4	69.4	68.2	61.0	48.6	37.3	31.9
		MCWB	28.3	27.7	30.2	33.8	43.5	53.7	60.5	59.1	52.0	42.5	34.0	29.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.6	33.7	34.6	42.2	52.9	63.4	67.5	68.7	60.8	49.9	42.9	36.7
		MCDB	36.9	36.7	36.9	48.4	64.7	76.0	77.3	75.9	67.1	52.9	43.6	38.5
	2%	WB	32.0	31.9	33.0	38.4	49.6	59.3	64.9	64.8	57.6	47.4	39.3	33.6
		MCDB	33.9	34.4	35.2	44.3	60.7	70.7	74.8	72.4	65.1	51.2	40.6	35.5
	5%	WB	30.4	30.1	31.8	36.3	46.7	56.7	62.7	62.1	55.4	45.2	36.9	31.8
		MCDB	32.8	32.4	33.9	41.3	55.7	67.2	71.6	69.1	61.3	49.0	38.6	34.0
	10%	WB	28.1	27.9	30.5	34.7	44.1	54.3	60.3	60.0	53.6	43.5	34.0	29.1
		MCDB	31.1	30.7	33.0	38.8	51.4	63.6	68.2	66.4	59.1	47.5	36.5	31.5
Mean Daily Temperature Range	5% DB	MDBR	14.3	16.0	16.0	13.5	16.2	17.7	16.3	16.1	18.2	15.9	14.4	14.6
		MCDBR	10.6	10.3	13.3	17.3	25.6	28.0	25.3	23.6	23.4	18.9	13.7	12.3
		MCWBR	9.8	8.5	9.9	11.4	14.9	14.8	13.3	12.7	13.0	11.6	11.1	11.3
	5% WB	MCDBR	11.9	9.6	10.5	15.2	23.6	25.5	22.7	20.5	19.2	15.6	11.8	12.3
		MCWBR	11.1	8.5	8.7	10.5	14.4	14.6	13.3	12.6	11.8	11.1	11.1	11.5
Clear-Sky Solar Irradiance	taub	0.307	0.341	0.403	0.462	0.515	0.492	0.495	0.454	0.380	0.367	0.326	0.300	
	taud	2.050	1.997	1.845	1.754	1.757	1.969	2.011	2.132	2.324	2.192	2.098	2.040	
	Ebn at Noon	198	229	237	239	234	243	243	239	243	250	221	188	174
	Edh at Noon	24	35	51	63	67	54	51	43	31	29	23	20	
All-Sky Solar Radiation	RadAvg	257	500	889	1350	1530	1613	1536	1285	1034	627	332	207	
	RadStd	32	74	91	152	150	192	142	128	94	63	33	29	